AI-Based Pneumonia Detection System with Power BI Dashboard

Phase 1 — Introduction & Concept Building

HEALTHCARETECHNOLOGY · COMPUTERSCIENCE

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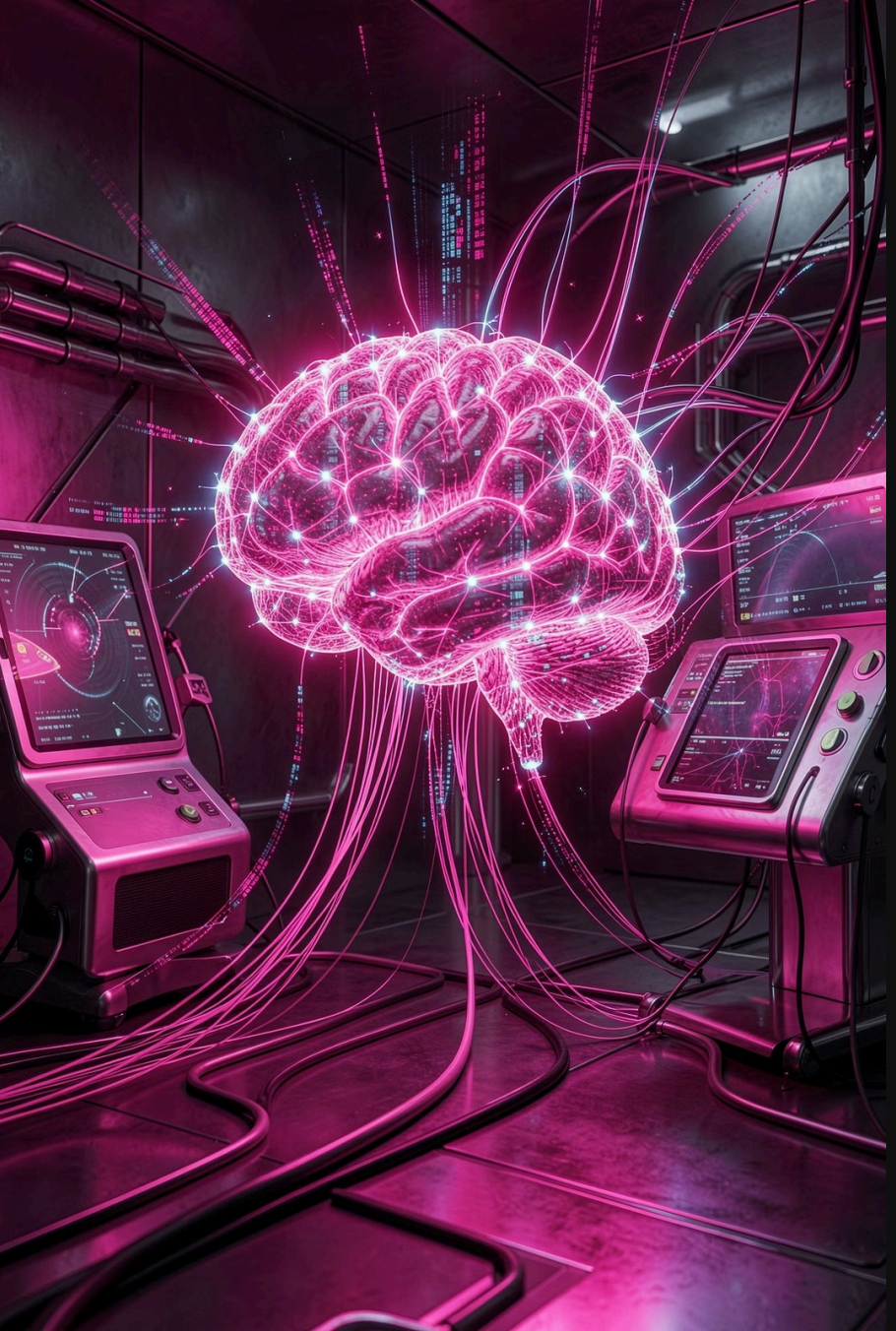


Artificial Intelligence in Healthcare

Artificial Intelligence (AI) is transforming modernmedicineby enabling computers tolearn from dataand make intelligentdecisions — oftenfaster and more accurately than humans alone.

Diagnosis Support

AI assists doctors in identifying diseases from scans, images, and reports.

Predictive Analytics

Models predict patient risk levels before symptoms worsen.

Workflow Efficiency

Reduces manual workload and speeds up clinical decisions.

What is Pneumonia?

Pneumonia is an **acute respiratory infection** that inflames theair sacs (alveoli) in one or both lungs, causing them to fill with fluid or pus.

Common Causes

- Bacterial infection (*Streptococcus pneumoniae*)
- Viral infection (influenza, COVID-19)
- Fungal pathogens in immunocompromised patients

Key Symptoms

- Persistent cough, fever, and chills
- Difficulty breathing and chest pain
- Fatigue and low oxygen saturation

Problem Statement

Delayed Diagnosis

ManualreviewofchestX-rays is time-consuming and subject to radiologist fatigue, leading to missed or late detections.

Resource Scarcity

Low-incomeregionslacktrained radiologists, leaving thousands of patients undiagnosed each year.

No Visual Analytics

Hospitalsrarelyhave real-timedashboards to track pneumonia trends, slowing down public health responses.

Why AI in Disease Detection?

Traditionaldiagnosticapproaches rely heavilyonhumanexpertiseandphysicalinfrastructure.AI bridges the gap — making accurate detection scalable, fast, and accessible.

Speed

AI analyzes an X-ray in seconds, compared to hours in manual workflows.

Accuracy

Deep learning models achieve clinical-grade accuracy on medical imaging tasks.

Accessibility

Cloud-based AI tools can reach remote clinics with minimal hardware requirements.

Data Insights

Integrated dashboards provide actionable trends to healthcare administrators.

Objectives of the Project

01

Detect Pneumonia from X-Rays

Build an AI model that classifies chest X-ray images as **Normal** or **Pneumonia** with high accuracy.

02

Create a Power BI Dashboard

Visualize detection results, patient statistics, and trends through an interactive real-time dashboard.

03

Support Clinical Decision-Making

Provide doctors with an intuitive tool that highlights confidence scores and flagged cases.

04

Improve Healthcare Efficiency

Reduce diagnostic time and manual effort, especially in under-resourced healthcare settings.

Scope of the Project

Thisprojectfocuses onawell-defined problem with a practical, deployable solution. Phase 1 establishes the conceptual and technical foundation.

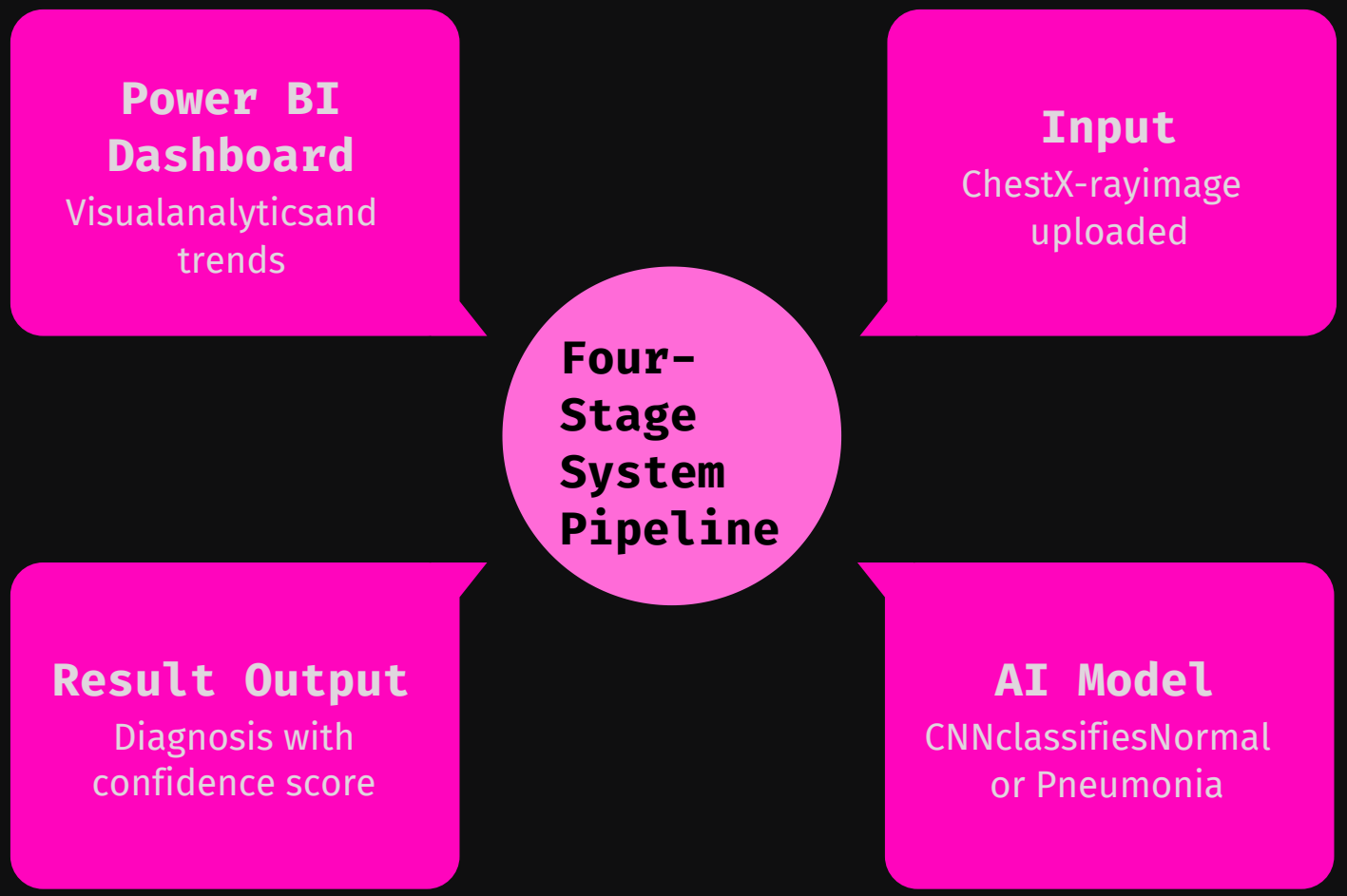
In Scope

- Chest X-ray image classification
- Binary output: Normal vs. Pneumonia
- Power BI integration for analytics
- Publicly available dataset (Kaggle)

Out of Scope

- Multi-disease classification
- Real-time hospital system integration
- Mobile application development
- Patient record management

Overview of the Proposed System



The system follows a straightforward pipeline — a chest X-ray is uploaded, processed by a trained Convolutional Neural Network (CNN), and the diagnosis result is fed into a Power BI dashboard for visualization and reporting.

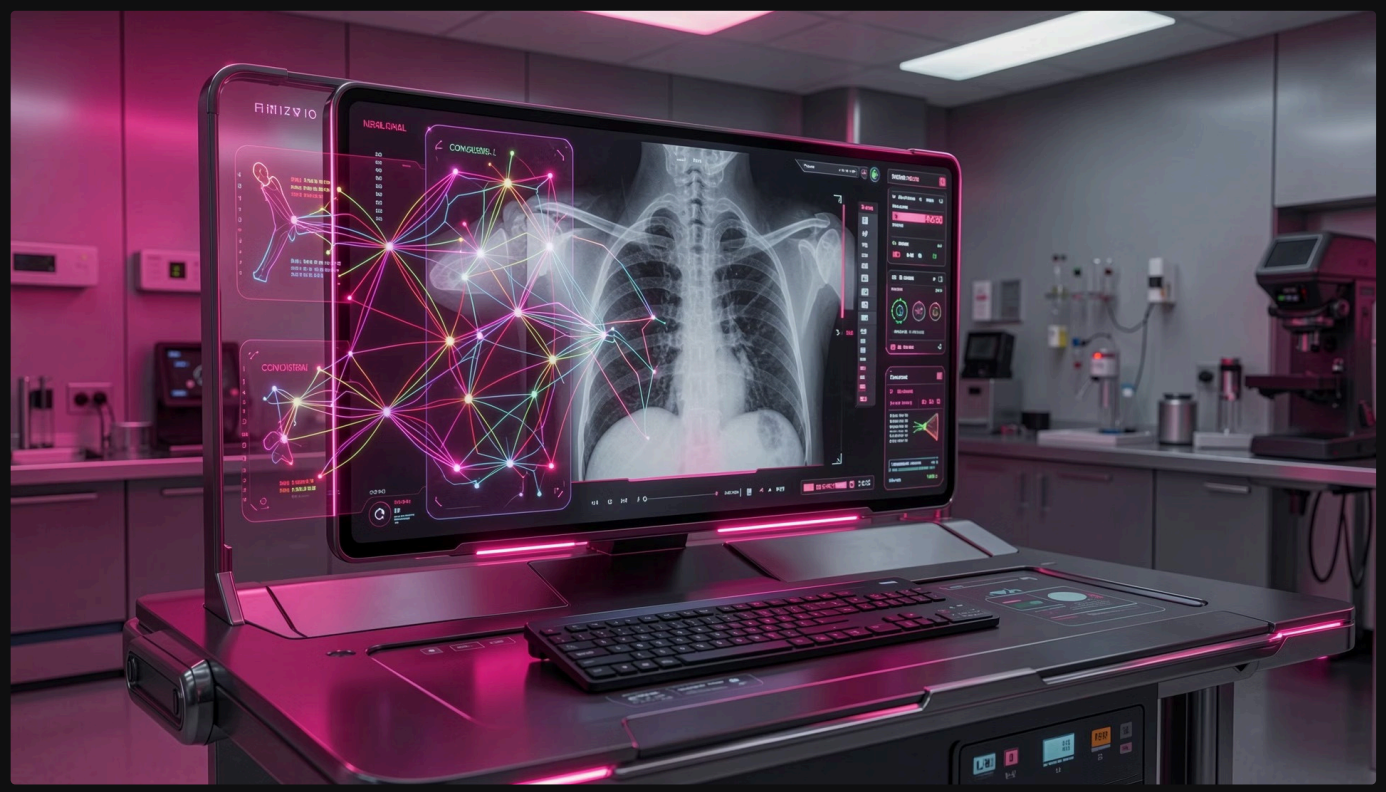
How AI Diagnoses from Images

Atthe coreof thissystemis a **Convolutional Neural Network (CNN)**— a type of deep learning model specifically designed to recognize patterns in images.

How It Works (Simply Put)

- The model is trained on thousands of labeled X-ray images
- It learns to identify visual patterns linked to pneumonia (e.g., lung opacity, fluid)
- When given a new X-ray, it predicts the class with a confidence score
- No manual feature extraction — the model learns on its own

> Think of it like teaching a child to distinguish cats from dogs — by showing many examples until the pattern is learned.



Phase 1 Summary

Phase1 lays theintellectualand structuralgroundwork for the entire project. Everysubsequentphase buildsdirectlyonthe understanding established here.

1

Problem Identified

Manual pneumonia detection is slow, error-prone, and inaccessible in many regions.

2

AI as the Solution

Deep learning (CNN) is selected as the core technology for X-ray image classification.

3

System Designed

A four-stage pipeline connects image input, AI inference, output, and Power BI reporting.

4

Ready for Phase 2

Dataset collection, model training, and dashboard development begin in the next phase.

The End

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